### 3.5.4 Attributes

An attribute is the information about that entity and it will describe, quantify, qualify, classify, and specify an entity. An attribute will always have a single value, that value can be a number or character or string.

**Types of attributes:**

1. Key Attribute
2. Simple Attributes
3. Composite Attributes
4. Single Valued Attribute
5. Multi Valued Attribute

#### 3.5.4.1 Key Attribute

Generally a key attribute describes a unique characteristic of an entity.

#### 3.5.4.2 Simple Attribute

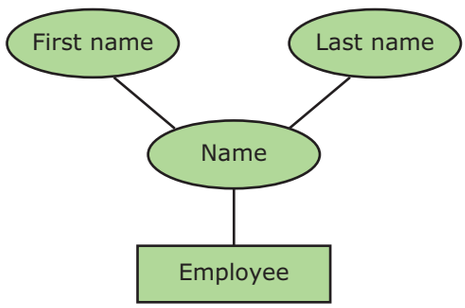



Figure 3.7 Simple Attribute

The simple attributes cannot be separated. It will have a single value for that entity. For Example: Let us consider the name as the attribute for the entity employee and here the value for that attribute is a single value. See Figure 3.7

#### 3.5.4.3 Composite Attributes

The composite attributes can be sub-divided into simple attributes without change in the meaning of that attribute. For Example: In the above diagram the employee is the entity with the composite attribute Name which are sub-divided into two simple attributes first and last name.

#### 3.5.4.4 Single Valued Attributes:

A single valued attribute contains only one value for the attribute and they don't have multiple number of values. For Example:Age- It is a single value for a person as we cannot give 'n' number of ages for a single person,therefore it is a single valued attribute. See Table 3.4

**Table: 3.4 Single Valued Attributes**

| Attribute | Values |
|---|---|
| Age | 3 |
| Roll no | 85 |

In the above table are the some examples for single valued attributes. See Figure 3.8

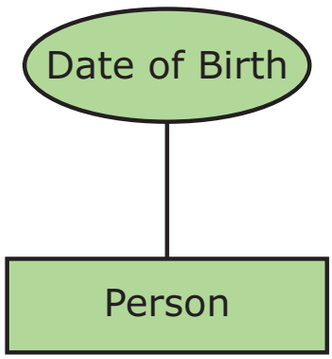



Figure 3.8 Single Valued Attributes

#### 3.5.4.5 Multi Valued Attributes:

A multi valued attribute has more than one value for that particular attribute.For Example:Degree - A person can hold n number of degrees so it is a multi-valued attribute.

In Table 3.5 are some examples for Multi valued attributes.

**Table: 3.5 Attributes and Values**

| Attribute | Values |
|---|---|
| Degree | B.Tech, MBA |
| Bank_Account | SBI, HDFC |



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